Chuanyang Jin

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Research Interests

Enhancing the robustness, truthfulness, and instruction-following capability of foundation models, such as large multimodal models and LLMs. Leveraging these models to build embodied agents capable of long-term decision making through planning, reasoning, managing memory, and learning from real-world feedback.

Education

New York University Sep 2020 – Expected May 2024

B.A in Computer Science with Honors; B.A in Mathematics *GPA: 3.96, Rank: 1/375*

- Relevant coursework: Deep Learning (Grad), Machine Learning (Grad), Natural Language Processing (Grad), Computer Vision (Grad), Algorithms (TA), Robot Intelligence, Differential Equations, Numerical Analysis

Research Experience

Massachusetts Institute of Technology May 2023 – Now

Visiting Researcher, advised by Joshua B. Tenenbaum, Antonio Torralba, Tianmin Shu *Boston*

New York University Feb 2022 – Now

Research Assistant, advised by Oded Regev, Saining Xie, He He *New York*

Publications / Preprints

- **C. Jin**, Y. Wu, J. Cao, J. Xiang, Y.-L. Kuo, Z. Hu, T. Ullman, A. Torralba, J. B. Tenenbaum, T. Shu. MMToM-QA: Multimodal Theory of Mind Question Answering. *In Submission to the 12th International Conference on Learning Representations **(ICLR)**, 2024.*
- K. Jha, T. A. Le, **C. Jin**, Y.-L. Kuo, J. B. Tenenbaum, T. Shu. Neural Amortized Inference for Nested Multi-agent Reasoning. *Accepted by the 38th AAAI Conference on Artificial Intelligence **(AAAI)**, 2024.*
- J. Wu*, Y. Xiao*, Y. Liu*, L. Wen, **C. Jin**, S. Liu, S. Paul, C. He, O. Regev, J. Fei. Deciphering RNA Localization in Nuclear Speckles. *In Submission to **Nature Communications**.*
- **C. Jin**. Self-Supervised Image Captioning with CLIP. ***NeurIPS**'23 Self-Supervised Learning Workshop, 2023.*
- P. Liu, **C. Jin**, Z. Zhong. Efficient Video Prediction and Segmentation with Sparse Labeled Data. *In Submission to **AAAI**'24 Artificial Intelligence with Biased or Scarce Data Workshop.*
- **C. Jin***, S. Zhang*, T. Shu, Z Cui. The Cultural Psychology of Large Language Models: Is ChatGPT a Holistic or Analytic Thinker? *In Submission to Proceedings of the National Academy of Sciences **(PNAS)**.*

Research Projects

MMToM-QA: Multimodal Theory of Mind Question Answering Jun 2023 – Oct 2023

Advisor: Prof. Joshua B. Tenenbaum, Prof. Antonio Torralba, Prof. Tianmin Shu

- Introduce MMToM-QA benchmark through procedural generation, which comprehensively evaluates the cognitive ability to understand people's minds (e.g., beliefs, goals) both on multimodal data and different unimodal data.
- Propose BIP-ALM (Bayesian Inverse Planning Accelerated by Language Models) to engineer multimodal ToM, which extracts symbolic representations from multimodal data, aligns and fuses representations, formulates an agent's behavior as a forward generative process, and finetunes language models to conduct inverse inference.
- Conduct a human experiment to verify our benchmark design and provide a natural performance yardstick.
- Demonstrate that current LLMs and multimodal models lack robust ToM capacity, and delve into the reasons.
- BIP-ALM shows promising results, leveraging the robustness of model-based planning and scalability of LLMs.
- Submit a paper (as the 1st author) to ICLR 2024.

Neural Amortized Inference for Nested Multi-agent Reasoning May 2023 – Aug 2023

Advisor: Prof. Joshua B. Tenenbaum, Prof. Tianmin Shu

- Introduce a novel method that recursively performs particle sampling using recognition networks and leveraging data from generative models to amortize high-order multi-agent reasoning and emulate human social inference.
- Design and construct two challenging multi-agent test scenarios in autonomous driving and construction domains.
- Experiments demonstrate that our method is computationally efficient and maintains exceptional accuracy.
- Publish a paper at AAAI 2024.

Optimization of Diffusion Transformers Feb 2023 – May 2023

Advisor: Prof. Saining Xie

- Implement optimizations such as pre-extracted VAE features, improved attention mechanisms, gradient checkpointing, and mixed precision training, achieving a 95% speed increase and 60% memory reduction.
- Investigate 4-bit post-training quantization of diffusion modes and progressive model distillation for fast sampling.
- Open-source the project, receiving over 250 GitHub Stars and subsequent integration into popular repositories.

Deciphering RNA Localization in Nuclear Speckles May 2022 – Oct 2023

Advisor: Prof. Oded Regev

- Analyze speckle-enriched transcripts and reveal sequence logics that drive RNA localization to nuclear speckles.
- Develop an extensive suite of tools for the analysis of experimental data, utilizing interpretable machine learning to decipher complex structures and interactions spanning over 30 factors, coupled with compelling visualizations.
- Uncover two sets of sequence features and propose a model where nuclear speckles kinetically coordinate splicing.
- Co-author a paper and present at the NYU Undergraduate Research Conference.

Visual Place Recognition Using Perspective Images Feb 2022 – May 2022

NYU AI4CE Lab

- Enhance a visual place recognition model to accommodate perspective images along with panoramas, by leveraging an encoder-decoder model to perform scene completion. Finetune it via a pretext task of relative pose estimation.
- Conduct experiments on simulated and real datasets, outperforming baselines in both recall rate and robustness.
- Develop a unified framework that handles various supervision methods, and both point cloud and RGB-D images.

Self-Supervised Image Captioning with CLIP May 2023 – Jul 2023

- Propose a self-supervised captioning method by maximizing the CLIP relevance between images and captions.
- Leverage CLIP encoding, mapping network, and LLaMA 2, bolstered by advanced initialization and optimization.
- Outperform SOTA methods utilizing only 2% of labeled data, measured by conventional and our novel metrics.
- Publish a paper (as the 1st author) at NeurIPS'23 Self-Supervised Learning Workshop.

Efficient Video Prediction and Segmentation with Sparse Labeled Data Apr 2023 – May 2023

- Propose a pseudo labeling and transfer learning method using a modified SimVP model for end-to-end video prediction and segmentation, effectively addressing the challenge of scarce labeled data and distribution shifts.
- Achieve substantial improvement over baseline methods in both Jaccard Index and MAE across multiple tasks.
- Submit a paper to AAAI'24 Artificial Intelligence with Biased or Scarce Data Workshop.

Work / Projects

Software Development Intern Jun 2021 – Aug 2021

Chinasoft International Ltd. *Nanjing*

- Maintain and extend the functionalities of a company's personnel/financial data management web platform.
- Design, implement, and test tools using MySQL to enable administrators to operate on shared databases.
- Leverage Django web framework, JavaScript, and HTML/CSS to create a user-friendly interactive interface.

AutoCorp: A Fully Autonomous Company Apr 2023 – May 2023

- Launch AutoCorp with LLMs as a core controller to generate reasoning traces and perform sequential actions.
- AutoCorp generates creative ads, branding, and product designs using GPT-4/DALL-E/ControlNet, interacts with customers, manages memory, learns from feedback, and performs long-term reasoning and automatic updates.

AutoLink: A Tool for Strategic Networking Mar 2023 – Apr 2023

- Develop a LinkedIn plugin integrating LLMs, data enrichment tools, graph-based search, and behavior analytics.
- AutoLink promotes strategic networking by profile information extraction, relevance-based ranking of potential connections/opportunities, personalized message generation, and adaptive learning from user preferences.

Class Embedding Patterns in Classifier-Free Diffusion Oct 2022 – Dec 2022

- Explore approaches to generate mappings from class labels and integrate them into the reverse diffusion process.
- Design and implement class label embedding patterns such as Uniform, Pyramid, and Bottleneck Embedding.
- Leverage curriculum learning to handle noisy labels. Evaluate their performance, properties, and trade-offs.

Parallel Command Processor Sep 2022 – Dec 2022

- Build a command interpreter that handles I/O redirection and pipe, and manages background/foreground process.
- Design a thread pool using POSIX threads to parallelize provided tasks, ensuring optimal system performance.
- Integrate a logging system to track command execution, errors, system events, and thread pool utilization.

Graph Visualization of Matrices Mar 2022 – May 2022

- Introduce a novel shift method to speed up the convergence of power iterations for computing tensor eigenpairs.
- Implement dimension reduction with power iterations and propose an innovative *graph Laplacian* concept for matrix visualization. Leverage them to create a toolbox that helps uncover structures in complex real-world relations.

Gene Mutation Detection System May 2019 – May 2021

- Propose a mutation detection algorithm for Sanger sequencing data, utilize computational geometry for feature extraction, employ deep learning for mutation localization and classification, and develop visualization tools.
- Launch an online platform adopted by leading hospitals to reduce diagnosis periods and receive a $10k reward.

Honors/Awards

- NYU Computer Science Prize for the Most Promising Junior Student (Top 1/375) 2023
- Finalist & MAA Award & COMAP Scholarship (Top 4/27205), Interdisciplinary Contest in Modeling 2022
- Champion Team, Tsinghua X-lab AI Hackathon 2021
- Distinguished Honor Roll (Top 1%), AMC12 & USAMO qualifier 2020
- International Bronze Award (Top 3/500), Shing-Tung Yau High School Computer Science Award 2019
- Finalist (Top 8/3790), FIRST Robotics Competition World Championship 2019
- First Prize (Top 0.1%), Chinese Mathematical Olympiad 2018
- Champion Team, International Regions Mathematics League 2018

Services/Activities

- Influential writer on Zhihu Q&A forum with 27K followers and 16,000K total reads 2020 – Present
- Teaching assistant of CSCI-UA 310 Basic Algorithms Fall 2022
- Executive board member of NYU Mathematics Society 2021 – 2022
- Director of Environmental Sustainability Committee at NYU Shanghai 2020 – 2021
- Activity Coordinator at World Wildlife Fund 2020 – 2021

Skills

- Language: Python, C++, C, Java, JavaScript, CSS, HTML, Matlab, SQL, R, Octave, Linux, Git
- Framework: PyTorch, TensorFlow, JAX, Scikit-learn, OpenCV, HuggingFace